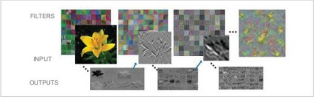

Figure 4: Image processing using deep learning

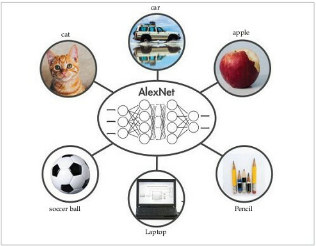

Figure 5: AlexNet

object is in each image. We label the images in order to have training data for the network.

Using this training data, the network can then start to understand the object's specific features and associate them with the corresponding category. Each layer in the network takes in data from the previous layer, transforms it, and passes it on. The network increases the complexity and detail of what it is learning from layer to layer. Notice that the network learns directly from the data—we have no influence over what features are being learned.

Implementation: An example using AlexNet

If you're new to deep learning, a quick and easy way to get started is to use an existing network, such as AlexNet, which is a CNN (convolutional neural network) trained on more than a million images. AlexNet is commonly used for image classification. It can classify images into 1000 different categories, including keyboards, computer mice, pencils, and other office equipment, as well as various breeds of dogs, cats, horses and other animals.

You will require the following software:

1. MATLAB 2016b or a higher version
2. Neural Network Toolbox
3. The support package for using Web cams in MATLAB (*https://in.mathworks.com/matlabcentral/fileexchange/45182-matlab-support-package-for-usb-webcams*)
4. The support package for using AlexNet (*https://in.mathworks.com/matlabcentral/fileexchange/59133-neural-network-toolbox-tm--model-for-alexnet-network*)

Figure 6: The Web cam's output

After loading AlexNet, connect to the Web cam and capture a live image.

Step 1: First, give the following commands:

```
camera = webcam; % Connect to the camera
nnet = AlexNet; % Load the neural net
picture = camera.snapshot; % Take a picture
```

Step 2: Next, resize the image to 227 x 227 pixels, the size required by AlexNet:

```
picture = imresize(picture,[227,227]); % Resize the picture
```

Step 3: AlexNet can now classify our image:

```
label = classify(nnet, picture); % Classify the picture
image(picture); % Show the picture
title(char(label)); % Show the label
```

Step 4: Give the following command to get the output:

```
output:
```

The table below lists a few differences between deep learning and machine learning

Machine learning	Deep learning
Good results with small data sets	Requires very large data sets
Quick to train a model	Computationally intensive
Needs to try different features and classifiers to achieve best results	Learns features and classifiers automatically
Accuracy plateaus	Accuracy is extremely high

END

References

[1] *file:///D:/deep%20learning/80879v00_Deep_Learning_ebook.pdf*
[2] *https://in.mathworks.com/help/nnet/examples/create-simple-deep-learning-network-for-classification.html*
[3] *https://in.mathworks.com/campaigns/products/offer/deep-learning.html*
[4] *https://in.mathworks.com/help/nnet/examples/transfer-learning-and-fine-tuning-of-convolutional-neural-networks.html*

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